

Assignment 1: Background

Part I: Fun with Probabilities, Utilities, and RL

(a)

Let F be the event that the fake coin is chosen, and N be the event that a normal coin is chosen.

$$P(F) = \frac{1}{n}, \quad P(N) = \frac{n-1}{n}$$

(i)

Let H_1 be the event of observing 1 head in 1 flip:

$$P(H_1 | F) = 1, \quad P(H_1 | N) = \frac{1}{2}$$

Applying Bayes' Theorem:

$$\begin{aligned} P(F | H_1) &= \frac{P(H_1 | F)P(F)}{P(H_1 | F)P(F) + P(H_1 | N)P(N)} \\ &= \frac{1 \cdot \frac{1}{n}}{1 \cdot \frac{1}{n} + \frac{1}{2} \cdot \frac{n-1}{n}} \\ &= \frac{\frac{1}{n}}{\frac{2+n-1}{2n}} = \frac{2}{n+1} \end{aligned}$$

(ii)

Let H_k be the event of observing k heads in k consecutive flips:

$$P(H_k | F) = 1^k = 1, \quad P(H_k | N) = \left(\frac{1}{2}\right)^k$$

Applying Bayes' Theorem:

$$\begin{aligned} P(F | H_k) &= \frac{P(H_k | F)P(F)}{P(H_k | F)P(F) + P(H_k | N)P(N)} \\ &= \frac{1 \cdot \frac{1}{n}}{1 \cdot \frac{1}{n} + \left(\frac{1}{2}\right)^k \cdot \frac{n-1}{n}} \\ &= \frac{\frac{1}{n}}{\frac{1}{n} \left(1 + \frac{n-1}{2^k}\right)} = \frac{2^k}{2^k + n - 1} \end{aligned}$$

(iii)

A false negative cannot occur because the fake coin has heads on both sides. An error occurs if and only if a normal coin is chosen and yields k consecutive heads:

$$\begin{aligned} P(\text{Error}) &= P(N \cap H_k) \\ &= P(N) \cdot P(H_k \mid N) \\ &= \frac{n-1}{n} \cdot \left(\frac{1}{2}\right)^k = \frac{n-1}{n \cdot 2^k} \end{aligned}$$

(b)

Let A, B, C denote the events that an ELT is manufactured by Altigauge, Bryant, and Chartair, respectively:

$$P(A) = 0.80, \quad P(B) = 0.15, \quad P(C) = 0.05$$

Let D denote the event that an ELT is defective:

$$P(D | A) = 0.04, \quad P(D | B) = 0.06, \quad P(D | C) = 0.09$$

(i)

$$P(B) = 0.15$$

(ii)

By Bayes' Theorem and the Law of Total Probability:

$$\begin{aligned} P(B | D) &= \frac{P(D | B)P(B)}{P(D | A)P(A) + P(D | B)P(B) + P(D | C)P(C)} \\ &= \frac{0.06 \cdot 0.15}{(0.04 \cdot 0.80) + (0.06 \cdot 0.15) + (0.09 \cdot 0.05)} \\ &= \frac{0.009}{0.032 + 0.009 + 0.0045} = \frac{0.009}{0.0455} = \frac{18}{91} \approx 0.1978 \end{aligned}$$

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(c)

The Axiom of Substitutability states that if an agent prefers lottery L_1 to L_2 ($L_1 \succ L_2$), then for any probability $p \in (0, 1]$ and any third outcome L_3 :

$$[p, L_1; (1 - p), L_3] \succ [p, L_2; (1 - p), L_3]$$

Let $L_3 = \$0$ with certainty and $p = 0.25$. We construct compound lotteries by mixing A and B with L_3 :

- **Mixing B with L_3 :**

$$[0.25, B; 0.75, \$0]$$

Since $B = \$3000$ with 100% probability, this yields:

$$\begin{aligned} 25\% \times 100\% &= 25\% \text{ chance of } \$3000 \\ 75\% &\text{ chance of } \$0 \end{aligned}$$

This compound lottery is equivalent to Lottery D.

- **Mixing A with L_3 :**

$$[0.25, A; 0.75, \$0]$$

Since $A = [80\% \text{ chance of } \$4000, 20\% \text{ chance of } \$0]$, this yields:

$$\begin{aligned} 25\% \times 80\% &= 20\% \text{ chance of } \$4000 \\ (25\% \times 20\%) + 75\% &= 80\% \text{ chance of } \$0 \end{aligned}$$

This compound lottery is equivalent to Lottery C.

Assuming an agent prefers $B \succ A$, the Axiom of Substitutability requires that mixing both sides with $L_3 = \$0$ at $p = 0.25$ must preserve the preference order:

$$[0.25, B; 0.75, \$0] \succ [0.25, A; 0.75, \$0] \implies D \succ C$$

However, choices in the Allais Paradox show that individuals choose $B \succ A$ and $C \succ D$. The preference $C \succ D$ directly contradicts the implied choice $D \succ C$, proving a violation of the Axiom of Substitutability.

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(d)

The utility function is given by $U(x) = -e^{-x/R}$.

(i)

For Mary, $U(x) = -e^{-x/500}$:

• **Option A:**

$$U(500) = -e^{-500/500} = -e^{-1} \approx -0.3679$$

• **Option B:**

$$\begin{aligned} E[U] &= 0.60 \cdot U(5000) + 0.40 \cdot U(0) \\ &= 0.60 \left(-e^{-5000/500} \right) + 0.40 \left(-e^0 \right) \\ &= 0.60 \left(-e^{-10} \right) - 0.40 \approx -0.4000 \end{aligned}$$

Since $U(500) > E[U]$ (i.e., $-0.3679 > -0.4000$), Mary chooses Option A.

(ii)

An individual is indifferent between Option 1 (\$100 certainty) and Option 2 (50% chance at \$500, 50% chance at \$0) when:

$$U(100) = 0.5 \cdot U(500) + 0.5 \cdot U(0)$$

Substituting $U(x) = -e^{-x/R}$:

$$\begin{aligned} -e^{-100/R} &= 0.5 \left(-e^{-500/R} - 1 \right) \\ 2e^{-100/R} &= e^{-500/R} + 1 \end{aligned}$$

Let $y = e^{-100/R}$. Then $e^{-500/R} = y^5$, giving the polynomial:

$$y^5 - 2y + 1 = 0 \implies (y - 1)(y^4 + y^3 + y^2 + y - 1) = 0$$

Solving $y^4 + y^3 + y^2 + y - 1 = 0$ gives $y \approx 0.5188$.

Solving for R :

$$e^{-100/R} = 0.5188 \implies -\frac{100}{R} = \ln(0.5188) \implies R \approx 152$$